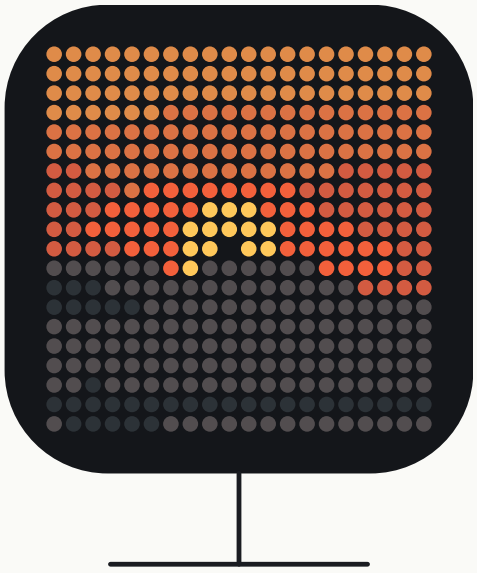
SPOTIPI

Photos — assembly guide



Your photos, and your album art, on 4,096 LEDs.
No soldering required. One power supply. No special tools.
About an hour to build, then a longer wait while software installs.
Works from a Windows PC or a Mac.

Read pages 2 and 3 before you plug anything in.

What's in front of you

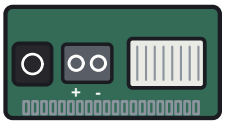
Lay everything out and check it off. Nothing here is supplied by us — you bought it yourself.



64x64 LED matrix panel, 3mm pitch

192 x 192 mm, and heavier than it looks (235 g).
Comes with its ribbon cable AND its power cable.
32x32 and 64x32 panels also work — see the guide.

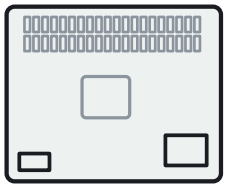
x 1



Adafruit RGB Matrix Bonnet

Arrives fully assembled — no soldering.
Not the HAT + RTC: that costs more and adds a clock this project never uses.

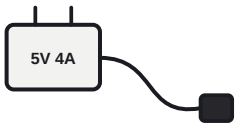
x 1



Raspberry Pi 3 Model A+

The 40 pins are already fitted.
Not a Pi Zero 2 W (out of stock, and marginal here), and not a Pi 5.

x 1



ONE 5V 4A power supply

2.1mm barrel plug, centre positive.
It powers the panel AND the Pi — the bonnet feeds the Pi through an onboard diode. See page 3.

x 1



microSD card

16-32GB. You'll also need a way to plug it into your computer.

x 1

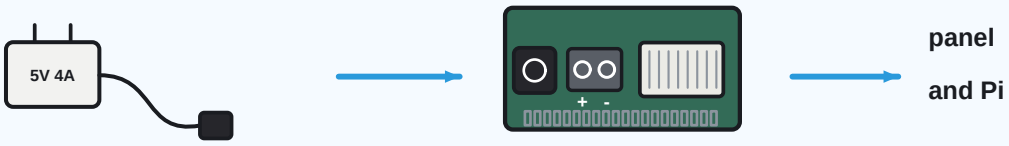
Tools needed

A small flat screwdriver, for two screw terminals. That is all for the standard build.
A soldering iron is optional, for the one-wire quality mod described on page 8.

Before you start

One power supply runs the whole thing

The 5V 4A supply plugs into the BONNET, and the bonnet feeds the Pi through a diode on the board. Adafruit put it plainly: "just plug in the 5V wall adapter into the bonnet and it will automagically power up the Pi too". You do not need a second supply for the Pi.



Add a second, micro-USB supply for the Pi ONLY if you run a 64x64 panel at brightness 90-100 and see the Pi's undervoltage warning. Most builds never do, and a 32x32 never will.

Two wires into the bonnet's screw terminal

RED	goes to	+	marked on the bonnet
BLACK	goes to	-	marked on the bonnet

Loosen the screw, insert the wire, tighten, then tug gently — neither should pull out.

Check both before the supply goes anywhere near a mains socket.

Two things to sort out while you wait for the parts

- **A free Spotify developer app.** Five minutes at developer.spotify.com/dashboard. It gives you three values the Pi needs later. No Premium required. The guide walks through it.
- **How photos will reach it.** Automatically from Apple Photos needs a Mac. From a Windows PC, a supplied script crops and copies them across whenever you want a change.

It is brighter than you are expecting

At full brightness this lights a whole room. Don't judge it on the first switch-on — you set brightness, a night dimmer and an off-schedule from your phone later.

1 Prepare the memory card

On your own computer — Windows or Mac. Nothing is plugged in yet.



Install Raspberry Pi Imager from raspberrypi.com/software, choose Raspberry Pi 3, and pick Raspberry Pi OS Lite (64-bit).

In Imager, click Edit Settings and fill in all of this:

- **Hostname** — spotipi
- **Username** — pi (and a password you write down)
- **Wi-Fi** — your network name and password, country GB
- **Time zone** — Europe/London — the night dimmer and off-schedule use it
- **Services tab** — tick Enable SSH, password authentication

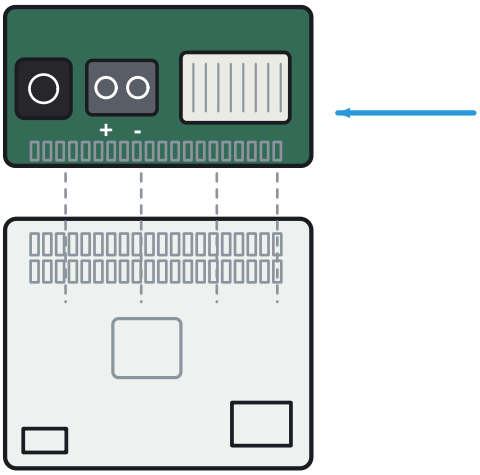
Then Save, Write, and wait. This erases the card — check you picked the card, not your computer's disk.

Why there is no keyboard or monitor in this guide

Those five settings are what let the Pi join your Wi-Fi and accept instructions on first boot. Get them right and you will never plug a screen into it. Get the Wi-Fi wrong and it is the one mistake that means starting this page again.

2 Push the bonnet onto the Pi

Everything unplugged. Line the pins up, press straight down.



all 40 pins engaged

Pi underneath, bonnet on top

Right

Every pin in a socket. Boards parallel, no gap at one end. Pressed home evenly, both hands.

It only fits one way round.

Wrong

Offset by one row, or a pin hanging over the end.

Don't force it. Lift it off, line it up again, and try once more.

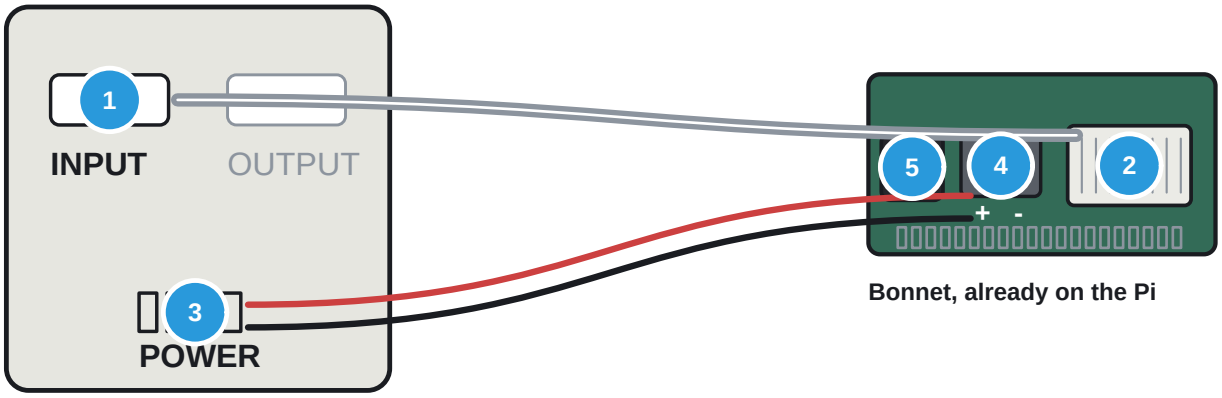
Think about where it will live, now rather than later

The ribbon cable in the next step is only about 16 cm long, so the Pi has to sit close behind or beside the panel — you cannot tuck it away on a shelf below.

The panel is 192 x 192 x 14 mm and 235 g, with M3 threaded holes on the back. It is too deep for a standard picture frame and too heavy for a picture pin. A deep box frame, or M3 standoffs onto a backing board, are the two arrangements that work.

3 Connect the panel

Two cables, both supplied with the panel. Nothing plugged into the mains yet.



BACK of the panel

Bonnet, already on the Pi

1	**Panel INPUT socket**	The grey ribbon goes here — not the socket beside it marked OUTPUT.
2	**Bonnet ribbon socket**	The other end of the same ribbon. Keyed: it only fits one way round.
3	**Panel POWER pins**	The four-hole plug on the power cable pushes on here.
4	**Bonnet screw terminal**	The other end: RED wire to +, BLACK wire to -. Tighten, then tug.
5	**Bonnet barrel jack**	The 5V 4A supply. Connect it last, and not at the wall yet.

If the sockets aren't clearly marked, the arrows on the board point AWAY from INPUT. Choosing wrong damages nothing — the panel simply stays dark — so if it does, swap to the other socket and try again.



The power cable that came with the panel: a four-hole plug at one end, two wires at the other.

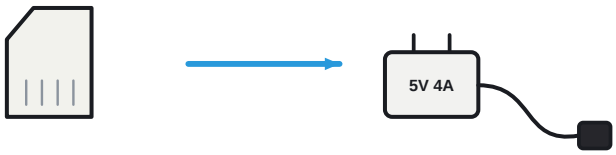
Red is +. Black is -.

Before you go on

Check the ribbon is square at both ends and neither wire pulls out. A half-seated ribbon looks like a software fault.

4 Card in, then switch on

One supply, one switch. Nothing to sequence.



microSD into the Pi

card slot on the underside

Switch on at the wall

panel and Pi both come up together

Wait 2-3 minutes for the first boot.

The panel will show nothing useful yet — the software isn't on it. That comes next, and it happens over the network from your computer.

Normal

The panel flickers, shows random colour noise, or stays dark.

A green light on the Pi, flickering irregularly.

Nothing is broken.

Stop

Anything hot to the touch.

Any smell of burning.

Pull the plug and re-check the red-to-plus, black-to-minus wiring on page 3.

The panel stays dark until the software is installed

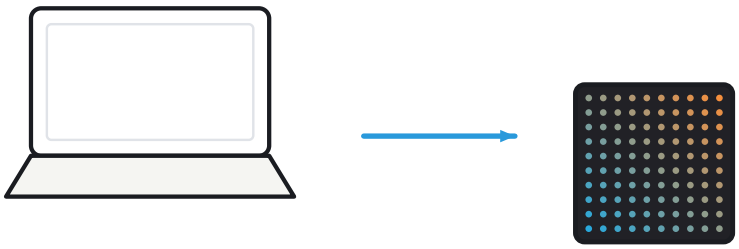
There is nothing on the card yet but the operating system, so an unlit panel now tells you nothing either way. Don't start rewiring.

What IS worth checking is that the Pi joined your Wi-Fi. On your computer open PowerShell (Windows) or Terminal (Mac) and run `ping spotipi.local` — replies mean page 4 was right and page 8 can begin.

No replies is almost always the Wi-Fi name or password. Re-flash the card and try again.

5 The rest happens on screen

The hardware is finished. The software takes about two hours, most of it waiting.



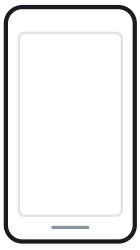
From a Windows PC or a Mac, over your own network. You never plug a keyboard or monitor into the Pi.

What's left, in order — all of it in the beginner's guide:

- **Spotify** — create a free developer app — Client ID, Secret and Redirect URI
- **Connect** — ssh pi@spotipi.local from PowerShell or Terminal
- **Panel driver** — Adafruit's installer, then a 10-20 minute compile
- **Token** — bash generate-token.sh — the one-off Spotify login
- **Install** — sudo bash install_pi.sh — sets up both background services
- **Photos** — an Apple Photos album on a Mac, or prepare_photos.py on any PC

Optional, and worth it: the one-wire quality mod

Solder one wire between GPIO 4 and GPIO 18 on the bonnet and the panel goes visibly steadier. Five minutes, one joint, reversible. The guide explains why, and how to switch it on afterwards.



Then it is all a web page

http://spotipi.local from your phone, on the same Wi-Fi.

Brightness, slideshow speed, a night dimmer, an off-schedule, and a live view of what is on the panel right now. Read SPOTIPI_BEGINNERS_GUIDE.md next.